

ICT600 CORRECTED STUDENT EDITION

December 2019 Corrected Student Scheme

The previously missing final-examination reference

Scope	December 2019 final examination
Evidence	Printed question, supplied scheme where available, current lecture notes, and technical verification
Status	Reconstructed from the printed paper and current lecture notes; no official or generated scheme was supplied

Prepared 26 September 2026

How to use this reference

Teaching-use document

Attempt the original question before revealing this scheme. This corrected edition is not an official UiTM replacement; it records known source errors and accepts equivalent technically correct answers.

- Assessment total: 100 marks: Part A 70, Part B 30.
- Model answers show one defensible solution; alternative correct wording or code may receive full credit.
- For code, method marks remain available when the approach is sound and a minor syntax error does not invalidate the solution.
- Security improvements are shown where historical examples trusted or displayed unvalidated input.

Question 1(a)

Describe Web Application, Web Server, and Web Browser.

[6 marks]

Model answer

- Web application: software accessed through a browser over a network, with processing on the client, server, or both.
- Web server: software/system that receives HTTP requests and returns static or generated web responses.
- Web browser: client software that requests web resources, renders HTML/CSS, and executes client-side JavaScript.

Marking guide: 2 marks per accurate definition: core identity (1) and function/distinction (1).

Question 1(b)

List FOUR types of cloud services.

[4 marks]

Model answer

- The current Chapter 1 note explicitly lists SaaS, IaaS, and PaaS.
- For the fourth item, accept another correctly explained service category such as FaaS/serverless or storage as a service; alternatively accept a clearly labelled cloud deployment model if that was the historical teaching context.

Marking guide: 1 mark per valid, consistently categorised item; maximum 4.

Correction / moderation note

The current lecture note contains only three service models, while the historical paper requests four. Do not invent a fourth from a different category without labelling the distinction.

Question 2(a)

Explain TWO target-attribute values for HTML links.

[4 marks]

Model answer

- `_self` opens the linked document in the current browsing context.
- `_blank` opens it in a new tab or window, depending on browser settings.
- Also accept `_parent` and `_top` with accurate explanations.

Marking guide: Any two: valid value (1) + correct effect (1).

Question 2(b)

Name SIX components of a website.

[6 marks]

Model answer

- Containing block, logo, navigation, content, footer, and whitespace.

Marking guide: 1 mark per correct component.

Question 3(a)

State TWO JavaScript variable-naming rules.

[4 marks]

Model answer

- A name may start with a letter, underscore, or dollar sign, but not a digit.
- Names are case-sensitive and cannot use reserved keywords.
- Meaningful camelCase names are good practice, though style alone is not a syntax rule.

Marking guide: Any two valid rules with brief explanation: 2 marks each.

Question 3(b)

Describe TWO ways to include jQuery in a web page.

[6 marks]

Model answer

- Download the library and reference the local file with a script element.
- Reference a trusted CDN URL with a script element; production use should include an integrity strategy and fallback where appropriate.

Marking guide: Method (1) + correct inclusion/explanation (2), for each of two methods.

Correction / moderation note

jQuery is historically relevant to this paper. Current projects may use modern native JavaScript when the library is not needed.

Question 4(a)

Produce the output of the supplied while and do-while script.

[6 marks]

Model answer

- with while: (no number follows because $1 < 0$ is false before the first iteration)
- with do-while: 1 (the body runs once before the false condition is checked)

Marking guide: Headings and control-flow explanation/output: 3 marks each.

Question 4(b)

Prepare a two-argument PHP function that produces the three printed person/year lines.

[6 marks]

Model answer

- One valid interpretation is to pass the first name and year while keeping Ahmed as the fixed surname.

```
<?php
function showPerson(string $firstName, int $year): void {
    echo $firstName . ' Ahmed. Born in ' . $year . '<br>';
}
showPerson('Maslina', 1989);
showPerson('Hardy', 1990);
showPerson('Azmi', 1987);
```

?>

Marking guide: Function/parameters (2), correct concatenation/output (2), three valid calls/results (2).

Correction / moderation note

The stem calls the first argument surname, but the sample varies the first name and fixes Ahmed. Mark a coherent implementation that reproduces the required output.

Question 4(c)

Loop through Axia, MyVi, and Bezza using an indexed PHP array.

[8 marks]

Model answer

- Create the indexed array, iterate through all three values, and display each value.

```
<?php
$scars = ['Axia', 'MyVi', 'Bezza'];
foreach ($scars as $scar) {
    echo htmlspecialchars($scar, ENT_QUOTES, 'UTF-8') . '<br>';
}
?>
```

Marking guide: Array (3), loop (3), complete display (2).

Question 5(a)

Create the illustrated Profil Pak Belang page.

[5 marks]

Model answer

- Use semantic headings/paragraphs, the supplied image, and a two-column table containing the three extinct subspecies.

```
<!doctype html>
<html lang="ms">
<head><meta charset="utf-8"><title>Profil Pak Belang</title></head>
<body>
<article>
  <h1>Profil Pak Belang</h1>
  <p><strong>Harimau Malaya</strong> (<em>Panthera tigris jacksoni</em>)
    merupakan salah satu subspecies harimau belang...</p>
  
  <p>Terdapat sembilan subspecies harimau belang...</p>
  <table border="1">
    <tr><th>Spesies</th><th>Nama Saintifik</th></tr>
    <tr><td>Harimau Bali</td><td><em>Panthera tigris balica</em></td></tr>
    <tr><td>Harimau Jawa</td><td><em>Panthera tigris sondaica</em></td></tr>
    <tr><td>Harimau Kaspian</td><td><em>Panthera tigris virgata</em></td></tr>
  </table>
  <p>Source: https://harimau.my/kenali-harimau-malaya/</p>
</article>
</body>
</html>
```

Marking guide: Valid structure (1), text formatting (1), image (1), complete table (1), source/content completeness (1).

Question 5(b)

Illustrate the page rendered by the supplied seminar-registration HTML.

[5 marks]

Model answer

- The rendered page contains a Seminar Registration Form heading; a Participant Registration Form fieldset; controls for name, email, gender, date of birth, institution, and category; and Register Me and Cancel submit controls.
- A labelled wireframe or accurate browser sketch receives credit; HTML source is not required because the question asks for the rendered result.

Marking guide: Heading/fieldset (1), core text controls (1), gender/date controls (1), category menu (1), action buttons/layout (1).

Question 5(c)

Create an online converter from MYR to USD at MYR 4.12 for USD 1.

[10 marks]

Model answer

- Because USD 1 costs MYR 4.12, divide the MYR amount by 4.12.

```
<!doctype html>
<html lang="en">
<body>
  <h1>Online Currency Converter from MYR to USD</h1>
  <label>MYR <input id="myr" type="number" min="0" step="0.01"></label>
  <button type="button" id="convert">Convert to USD</button>
  <script>
    document.getElementById('convert').addEventListener('click', () => {
      const myr = Number(document.getElementById('myr').value);
      if (!Number.isFinite(myr) || myr < 0) {
        alert('Enter a valid MYR amount.');
```

Marking guide: HTML input/button (3), event/function (2), validation (1), correct division (2), formatted alert/result (2).

Correction / moderation note

The printed screenshot shows MYR 5 = USD 20.6, which multiplies and therefore performs the opposite conversion. The corrected answer follows the written MYR-to-USD requirement.

Part B Question 1

Design an e-commerce product-page wireframe with EIGHT features.

[8 marks]

Model answer

- Any eight clearly labelled features, such as logo/header, search, navigation, breadcrumb, product image gallery, product name/price, description/specifications, options/quantity, add-to-cart action, stock/delivery details, ratings/reviews, related products, or footer.

Marking guide: 1 mark per relevant, visible and logically placed feature; maximum 8.

Part B Question 2

Create the PHP login flow shown in the wireframes, with client-side email validation and session variables.

[22 marks]

Model answer

- The model uses an email control for browser-side validation, POST handling, initial Guest session values, credential checking, an error state, and a welcome state.
- Equivalent multi-file login and welcome pages are acceptable.

```
<?php
session_start();
$_SESSION['username'] ??= 'Guest';
$_SESSION['role'] ??= 'Guest';
$error = '';

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $email = trim($_POST['email'] ?? '');
    $password = $_POST['password'] ?? '';
    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        $error = 'Enter a valid email address.';
    } elseif ($email === 'amran@mydomain.com' && $password === 'Sec@1ret') {
        session_regenerate_id(true);
        $_SESSION['username'] = 'amran';
        $_SESSION['role'] = 'member';
    } else {
```

```

        $error = 'Incorrect username and/or password. Please login again.';
    }
}
function e(string $v): string { return htmlspecialchars($v, ENT_QUOTES, 'UTF-8'); }
?>
<!doctype html><html lang="en"><head><meta charset="utf-8"><title>Login</title></head>
<body>
<?php if ($_SESSION['role'] === 'member'): ?>
    <h1>Welcome</h1><p>Hello <?= e($_SESSION['username']) ?></p>
    <p>User role as <?= e($_SESSION['role']) ?></p>
<?php else: ?>
    <h1>Login</h1><?php if ($error): ?><p><?= e($error) ?></p><?php endif; ?>
    <form method="post">
        <label>Username <input type="email" name="email" required></label>
        <label>Password <input type="password" name="password" required></label>
        <button type="submit">Login</button>
    </form>
<?php endif; ?>
</body></html>

```

Marking guide: Form/email/password (5), validation/request handling (4), credential decision and messages (5), session start/default/update (5), safe output and coherent flow (3).

Correction / moderation note

The fixed password is retained only because it is specified by the historical exercise. Real systems must store password hashes and compare with password_verify.

Consistency rules

- Award equivalent technically correct syntax and wording.
- Do not reproduce a known historical error merely because it appears in a supplied scheme.
- Keep printed subpart totals fixed unless a documented moderation decision says otherwise.
- For scenario questions, connect the explanation or mitigation to the scenario.
- Record additional accepted alternatives for the next controlled revision.

December 2019 - Complete Coding Answers

Exam-schema presentation

Each answer below includes executable model code and an awardable marking schedule that sums to the printed marks. Equivalent working solutions receive the same marks when they satisfy the required behaviour.

Question 4(a) - PHP loop trace

6 marks

Produce the exact output of the supplied while and do-while code.

Mark allocation

Awardable element	Marks
First heading	1
Correctly shows no while-loop number	2
Second heading	1
Correct do-while value	1
Explains pre-test versus post-test behaviour	1
Total	6

Complete model answer

```
with while:  
(no number is printed)  
with do-while:  
1
```

Question 4(b) - PHP function

6 marks

Write the complete function and calls that reproduce all three required lines.

Mark allocation

Awardable element	Marks
Function and two parameters	2
Correct concatenated output	2
Three correct calls	2
Total	6

Complete model answer

```
<?php  
function showPerson(string $firstName, int $year): void {  
    echo $firstName . ' Ahmed. Born in ' . $year . '<br>';  
}  
showPerson('Maslina', 1989);  
showPerson('Hardy', 1990);  
showPerson('Azmi', 1987);  
?>
```

Marker note

The stem says surname, but the examples vary the first name and keep Ahmed fixed. Accept any coherent implementation that reproduces the three lines.

Question 4(c) - indexed array loop

8 marks

Use an indexed PHP array and display Axia, MyVi and Bezza.

Mark allocation

Awardable element	Marks
Correct indexed array	3
Correct loop	3
All three values displayed	2
Total	8

Complete model answer

```
<?php
$scars = ['Axia', 'MyVi', 'Bezza'];
foreach ($scars as $scar) {
    echo $scar . '<br>';
}
?>
```

Question 5(a) - Profil Pak Belang page

5 marks

Create the complete HTML page shown in Figure 1.

Mark allocation

Awardable element	Marks
Valid page and heading/paragraph structure	1
Required text formatting	1
pak-belang.png image	1
Complete two-column table	1
Source and content completeness	1
Total	5

Complete model answer

```
<!doctype html>
<html lang="ms">
<head><meta charset="utf-8"><title>Profil Pak Belang</title></head>
<body>
<article>
<h1>Profil Pak Belang</h1>
<p><strong>Harimau Malaya</strong> (<em>Panthera tigris jacksoni</em>)
    merupakan salah satu subspecies harimau belang...</p>

<p>Terdapat sembilan subspecies harimau belang...</p>
<table border="1">
    <tr><th>Spesies</th><th>Nama Saintifik</th></tr>
    <tr><td>Harimau Bali</td><td><em>Panthera tigris balica</em></td></tr>
    <tr><td>Harimau Jawa</td><td><em>Panthera tigris sondaica</em></td></tr>
    <tr><td>Harimau Kaspian</td><td><em>Panthera tigris virgata</em></td></tr>
</table>

<p>Source: <a href="https://harimau.my/kenali-harimau-malaya/">https://harimau.my/kenali-harimau-malaya/</a></p>
</article>
</body>
</html>
```

Question 5(c) - MYR to USD converter

10 marks

Create the full HTML and JavaScript converter at MYR 4.12 for USD 1.

Mark allocation

Awardable element	Marks
Input, label and conversion button	3
Event handler/function	2
Numeric validation	1
Correct MYR divided by 4.12	2
Formatted result	2
Total	10

Complete model answer

```
<!doctype html>
<html lang="en">
<body>
  <h1>Online Currency Converter from MYR to USD</h1>
  <label>MYR <input id="myr" type="number" min="0" step="0.01"></label>
  <button type="button" id="convert">Convert to USD</button>
  <script>
    document.getElementById('convert').addEventListener('click', () => {
      const myr = Number(document.getElementById('myr').value);
      if (!Number.isFinite(myr) || myr < 0) {
        alert('Enter a valid MYR amount.');
```

```
      return;
    }
    const usd = myr / 4.12;
    alert(`MYR ${myr.toFixed(2)} = USD ${usd.toFixed(2)}`);
  });
</script>
</body>
</html>
```

Expected output / behaviour

MYR 5.00 = USD 1.21

Marker note

The historical screenshot multiplies by 4.12 and therefore performs the opposite conversion. The written MYR-to-USD requirement must divide.

Part B Question 2 - PHP login and session flow

22 marks

Provide a complete runnable login solution matching the required guest, error and welcome states.

Mark allocation

Awardable element	Marks
Login form and email/password controls	5
Request and email validation	4
Credential decision and messages	5
Session start, default values and authenticated update	5
Safe output and coherent page flow	3
Total	22

Complete model answer

```
<?php
session_start();
```

```

$_SESSION['username'] ??= 'Guest';
$_SESSION['role'] ??= 'Guest';
$error = '';

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $email = trim($_POST['email'] ?? '');
    $password = $_POST['password'] ?? '';
    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        $error = 'Enter a valid email address.';
    } elseif ($email === 'amran@mydomain.com' && $password === 'Sec@1ret') {
        session_regenerate_id(true);
        $_SESSION['username'] = 'amran';
        $_SESSION['role'] = 'member';
    } else {
        $error = 'Incorrect username and/or password. Please login again.';
    }
}

function e(string $v): string { return htmlspecialchars($v, ENT_QUOTES, 'UTF-8'); }
?>
<!doctype html><html lang="en"><head><meta charset="utf-8"><title>Login</title></head>
<body>
<?php if ($_SESSION['role'] === 'member'): ?>
    <h1>Welcome</h1><p>Hello <?= e($_SESSION['username']) ?></p>
    <p>User role as <?= e($_SESSION['role']) ?></p>
<?php else: ?>
    <h1>Login</h1><?php if ($error): ?><p><?= e($error) ?></p><?php endif; ?>
    <form method="post">
        <label>Username <input type="email" name="email" required></label>
        <label>Password <input type="password" name="password" required></label>
        <button type="submit">Login</button>
    </form>
<?php endif; ?>
</body></html>

```

Marker note

A fixed exercise password is retained only to match the historical paper. Real applications must store password hashes and use `password_verify()`.

Coding-marker consistency

- Award method marks when the core algorithm and required structure are correct.
- Do not double-penalise one syntax error that causes several later lines to fail.
- Accept equivalent DOM methods, loop forms, variable names, SQL aliases, and control structures.
- Require every behaviour named in the question; a prose description alone is not a complete coding answer.
- For unsafe historical code, award course-aligned logic while teaching prepared SQL, password hashing, validation, and output encoding.